The ELSA Architecture: A Framework for a Co-Evolutionary, Self-Adaptive Linguistic System

Part I: Foundational Architecture - From Monolithic Prompt to Multi-Agent Swarm

The prevailing paradigm of large language model (LLM) interaction relies on monolithic prompt architectures. While capable of impressive feats, these single, complex instruction sets are inherently brittle. They represent a single point of failure, are notoriously difficult to debug and iterate upon, and lack the intrinsic adaptability required for robust performance in dynamic, open-ended environments. Complex cognitive tasks, much like in human endeavors, benefit from a diversity of perspectives, collaborative refinement, and specialized roles—features that are structurally precluded by a monolithic design. To overcome these limitations, this report proposes a fundamental architectural update for the Emergent Linguistic Systems Architect (ELSA): a transition from a static, singular prompt to a dynamic, decentralized collective of specialized agents, organized as a multi-agent swarm. This foundational shift is not merely an enhancement but a necessary evolution to unlock higher-order capabilities such as autonomous adaptation, emergent problem-solving, and verifiable safety.

1.1. The Limitations of Monolithic Architectures

A monolithic prompt, regardless of its complexity or cleverness, functions as a single, centralized controller. This design choice introduces several critical vulnerabilities. Firstly, it creates a single, high-stakes attack surface; a minor, adversarially crafted perturbation can cause catastrophic failure across the entire system. Secondly, the interconnectedness of instructions within a large prompt makes iterative improvement challenging. A change intended to improve one aspect of performance can have unforeseen and detrimental effects

on others, leading to a cumbersome and unpredictable development cycle. Finally, and most importantly, a monolithic structure cannot effectively model the multi-faceted reasoning processes required for truly complex problems. It forces a single "mind" to simultaneously embody the roles of generator, critic, reasoner, and planner, a cognitive load that inevitably leads to compromises in the quality of each function. This section deconstructs this outdated paradigm to establish the architectural necessity for a more resilient, scalable, and collaborative foundation.

1.2. Introducing the Multi-Agent System (MAS) Paradigm for ELSA

The proposed ELSA architecture is reconceptualized as a Multi-Agent System (MAS), a computational framework comprising multiple interacting, intelligent agents.[1] Within this paradigm, ELSA is not a single entity but a society of LLM-based agents, where each agent is an independent unit of behavior with a unique profile, role, and operational logic.[3] This architectural shift from a single model to a collaboration-centric approach enables the system to tackle complex tasks through the emergent synergy of specialized components, mirroring the efficacy of human teamwork.[1] The system's dynamics are governed by principles derived from the study of swarm intelligence, which provides a robust blueprint for achieving sophisticated collective behavior from simple, local interactions.[6]

Three core principles of swarm intelligence form the bedrock of the ELSA MAS architecture:

- **Decentralized Control:** The system operates without a central leader or master controller. Decision-making authority is distributed among the individual agents, each acting based on local information and its specific role.[7] This decentralization eliminates single points of failure, dramatically increasing the system's overall robustness and fault tolerance. If one agent fails or produces a flawed output, the rest of the swarm can identify, correct, or bypass the error, ensuring operational continuity.[6]
- **Self-Organization:** Complex, globally coherent behavior arises spontaneously from the local interactions between agents, without the need for explicit, top-down coordination.[6] Positive and negative feedback loops, driven by agent interactions with each other and their shared environment, allow the system to dynamically structure its problem-solving approach. This capacity for self-organization enables the swarm to adapt fluidly to novel tasks and changing environmental conditions without requiring external reprogramming.[7]
- **Emergent Behavior:** The collective intelligence and problem-solving capability of the swarm significantly surpass the abilities of any single constituent agent.[6] Non-linear interactions between agents can lead to the discovery of novel, unexpected, and highly effective solutions to complex problems. This emergent property is the ultimate goal of the MAS architecture, transforming the system from a mere task-executor into a genuine

problem-solving engine.[7]

The shift to a multi-agent swarm is not just an efficiency gain; it's a fundamental re-architecting for resilience. A diverse population of specialized agents creates a system that is inherently more robust against adversarial attacks or single-point failures. This decentralization functions as a form of innate immune response. The principles of swarm intelligence emphasize robustness and fault tolerance through redundancy and distributed control [6], directly countering the known vulnerabilities of AI systems to adversarial examples.[9] While an attack might successfully deceive a Generator agent, a well-designed Critic agent, operating with a different set of evaluation criteria, can flag the anomaly. This functional differentiation and distributed verification create a system far more resilient than any monolithic counterpart.

1.3. Defining Agent Profiles and Roles within the ELSA Swarm

A key advantage of the MAS paradigm is the ability to deploy heterogeneous agents with specialized profiles and capabilities.[1] The initial configuration of the ELSA swarm will consist of several distinct agent classes, each with a clearly defined role in the collective cognitive workflow:

- **Generator Agents:** These agents are responsible for the initial creation of content, solutions, or responses based on a given task. To foster diversity of thought, the system will employ multiple Generator agents in parallel, each with a distinct "persona" or cognitive style (e.g., a highly creative and divergent generator, a logically rigorous and analytical generator, a cautious and detail-oriented generator).
- **Critic Agents:** The function of Critic agents is to evaluate the outputs produced by the Generator agents. Each Critic is specialized, assessing outputs against a specific set of criteria, such as factual accuracy, logical consistency, adherence to constraints, ethical alignment, or stylistic coherence. This division of evaluative labor ensures a comprehensive and multi-faceted review process.
- **Refiner Agents:** These agents serve as the bridge between generation and finalization. They take the outputs from Generator agents, along with the feedback from Critic agents, and perform iterative improvements. Their goal is to synthesize the creative potential of the generators with the rigorous standards of the critics to produce a high-quality final product.
- **Orchestrator Agent:** This is a lightweight, specialized agent whose role is to manage the flow of information and tasks within the swarm. It routes initial requests to appropriate Generator agents, passes their outputs to the relevant Critics, and directs criticized content to Refiners. Crucially, the Orchestrator does not hold centralized control over the solution; its function is to facilitate communication and maintain an efficient workflow, not

to dictate outcomes.
- **Observer Agent:** This is a meta-agent that operates outside the primary task-solving loop. Its role is to monitor, record, and analyze the activities and evolution of the entire system. It is the core of ELSA's self-analysis capability, which will be detailed in Part IV.

1.4. Communication and Collaboration Protocols

Effective collaboration within the swarm depends on structured and efficient communication protocols. The choice of protocol is critical, as it directly influences the system's performance and its ability to avoid common MAS failure modes.[10] ELSA will employ a hybrid communication model that combines indirect environmental interaction with structured, direct dialogue.

- **Stigmergy (Indirect Communication):** The primary mode of coordination will be stigmergic. Agents will interact indirectly by modifying a shared, structured "workspace" or environment—for example, a JSON object or a dedicated database schema that represents the current state of the solution.[6] This method, inspired by ant colonies that coordinate via pheromone trails, is highly scalable and efficient for decentralized systems. It provides a persistent, objective record of the task's progress, which all agents can reference.[6] This structured workspace is a powerful defense against the semantic drift that plagues purely conversational systems. The problem of semantic drift arises when meaning erodes over successive turns in a dialogue as context is fluidly re-interpreted at each step.[13] A stigmergic workspace, by contrast, acts as an external, incorruptible memory. It provides a constant "ground truth" to which agents must continuously re-anchor their contributions, preventing the subtle but cumulative degradation of intent.
- **Structured Debate and Discussion Paradigms:** For tasks that require complex reasoning, negotiation, or the resolution of conflicting viewpoints, agents will engage in direct, structured debates. The architecture will support multiple debate paradigms, which can be selected based on the task requirements [10]:
  - **Memory Paradigm:** All participating agents have access to the complete history of the discussion, promoting the formation of a shared understanding and facilitating consensus.[10]
  - **Relay Paradigm:** Agents communicate in a sequential chain, where each agent only sees the output of the immediately preceding agent. This is useful for building linear chains of reasoning or creative narratives.[10]
  - **Debate Paradigm:** A formal, turn-based structure where agents present arguments, critiques, and rebuttals. This adversarial format is designed to rigorously test ideas and refine complex solutions.[10]
- **Consensus Mechanisms:** To ensure that discussions are productive and lead to a final

decision, the system will employ explicit consensus protocols. These mechanisms prevent endless loops and provide a clear termination condition for collaborative tasks. Options include Majority Consensus, where a decision is finalized once it receives support from over 50% of the agents, or the use of a specialized Judge agent that reviews the debate and makes a final determination.[10]

1.5. Addressing Inherent Challenges in Multi-Agent Conversational Systems

The design of the ELSA architecture proactively incorporates mitigations for known failure modes identified in recent MAS research. By anticipating these challenges, the system can be engineered for greater stability and reliability.

- **Problem Drift:** This phenomenon occurs when long, unstructured discussions cause agents to gradually deviate from the original task requirements, leading to irrelevant or incorrect final outputs.[17] ELSA mitigates this risk through the use of strictly defined, turn-limited debate protocols and clear consensus mechanisms that enforce a timely conclusion to discussions. The stigmergic workspace also serves as a constant anchor to the original goal.[10]
- **Alignment Collapse:** Research indicates that the ethical alignment of multi-agent systems can degrade over the course of prolonged, unguided interactions, potentially leading to the generation of unsafe or undesirable content.[17] This is addressed in two ways: first, by the explicit inclusion of specialized Critic agents whose function is to enforce ethical and safety constraints, and second, through the integration of a co-evolving safety system (detailed in Part V) that dynamically adapts to counter emergent misalignments.[10]
- **Monopolization:** In some conversational structures, a single agent can dominate the discussion, drowning out other perspectives and reducing the collective benefit of the multi-agent approach.[17] The Orchestrator agent, combined with turn-based debate protocols, mitigates this by ensuring that all relevant agents have a fair opportunity to contribute, thereby preserving the cognitive diversity of the swarm.[12]

Part II: The Evolutionary Engine - Prompt Co-evolution via Genetic Algorithms

The multi-agent architecture described in Part I provides a robust and flexible foundation. To transform this system from merely static to genuinely evolutionary, a mechanism for long-term, structural adaptation is required. This section details the ELSA Evolutionary Engine, which uses principles from Genetic Algorithms (GAs) to continuously optimize the entire population of prompt-agents. By treating the agent swarm as a gene pool, the system can explore a vast design space, discovering novel and more effective configurations over successive generations.

2.1. Genetic Representation of Prompt-Agents

To enable evolution, the core identity of each agent must be encoded in a format that can be manipulated by genetic operators. This genetic representation, or "chromosome," will be a structured data object (e.g., a YAML or JSON file) that encapsulates all defining aspects of an agent:

- **Prompt DNA:** The full text of the agent's system prompt, broken down into semantic blocks (e.g., core identity, goal, constraints, output format).
- **Parameter Genes:** Key operational parameters, such as the LLM's temperature, top-p, and presence penalties, encoded as numerical values.
- **Logic Alleles:** Categorical flags or references to specific sub-modules that define the agent's operational logic, such as its communication protocol or its role in a debate structure.

This comprehensive encoding allows the evolutionary process to modify not just the wording of a prompt but the fundamental behavior and configuration of the agent itself, enabling a much richer and more powerful evolutionary search.

2.2. The Genetic Operators: Mutation, Crossover, and Selection

The evolution of the agent population is driven by three primary genetic operators, analogous to those in biological evolution [19]:

- **Mutation:** This operator introduces novelty and drives exploration by making small, random changes to an agent's chromosome. A mutation might involve rephrasing a sentence in the prompt DNA, slightly altering a parameter gene, or switching a logic allele to an alternative setting (e.g., changing a Critic's evaluation metric from "conciseness" to "comprehensiveness").
- **Crossover (Recombination):** This operator facilitates the combination of successful

traits from different agents. Two high-performing "parent" agents are selected, and their chromosomes are combined to create one or more "offspring" agents. For example, the goal-defining section of one parent's prompt could be merged with the constraint-handling section of another, potentially creating a new agent that inherits the strengths of both.[19]
- **Selection:** This is the driving force of the evolutionary process. After evaluating the performance of all agents in a generation, a selection mechanism determines which agents will "reproduce" and pass their genetic material to the next generation. Agents with higher fitness scores have a greater probability of being selected, ensuring that effective traits are propagated and refined over time.

2.3. Designing a Multi-Objective Fitness Function

The fitness function is the most critical component of the GA, as it defines the selection pressures that shape the evolution of the swarm. A simplistic, single-objective fitness function (e.g., measuring only task accuracy) is a known failure mode for GAs, as it often leads to the evolution of "over-fit" or brittle solutions that perform well on one metric at the expense of all others.[19] To create a well-rounded and robust agent population, ELSA's fitness function will be a weighted composite of multiple, sometimes competing, objectives:

- **Performance:** A measure of the agent's (and the swarm's) success in completing its assigned tasks. This can be based on metrics like accuracy, quality scores, or user satisfaction ratings.
- **Efficiency:** A measure of the computational resources consumed. This includes factors like token usage, inference time, and the number of repair loops required to reach a solution. This creates pressure for more concise and efficient agents.
- **Robustness:** A measure of the agent's performance on out-of-distribution, noisy, or adversarial inputs. This is evaluated by the Guardian agents (Part V) and directly incentivizes the evolution of more resilient behaviors.
- **Semantic Fidelity:** A direct penalty for semantic drift, as measured by the Semantic Drift Protocol (Part IV). This is a crucial component for mitigating the risks of self-referential learning, where a system learning from its own outputs can enter a feedback loop of degrading quality and diversity.[22] This objective creates a strong evolutionary pressure to maintain the integrity of core concepts.
- **Collaboration Score:** A measure of an agent's positive contribution to the collective success of the swarm. This rewards agents that provide useful feedback, generate high-quality initial drafts, or otherwise facilitate the group's workflow, promoting the evolution of cooperative behaviors.

While evolutionary systems are often perceived as opaque "black boxes," the ELSA

architecture is designed to leverage its own evolutionary history as a powerful tool for interpretability. The Observer agent (detailed in Part IV) will maintain a comprehensive log of every genetic change and its corresponding impact on the multi-objective fitness score. Over thousands of generations, this ledger becomes a rich dataset. By applying statistical analysis to this data, it becomes possible to identify which "genes"—be they specific prompt phrases, parameter settings, or logical structures—consistently correlate with high performance. For instance, if agents whose prompts include the phrase "reason step-by-step from first principles" consistently achieve higher fitness scores across a diverse range of tasks, this provides strong, empirically-derived evidence for the causal importance of that instruction. This process transforms the seemingly random walk of evolution into a large-scale, automated ablation study, revealing the most critical components of the system's own internal logic and making its success more transparent and understandable.

2.4. Mitigating GA Failure Modes: Escaping Local Optima

A common failure mode for GAs is premature convergence, where the entire population becomes trapped in a "local optimum"—a good but not the best possible solution—because any small mutation results in a decrease in fitness.[19] To ensure that ELSA can navigate a complex fitness landscape and continue to find better solutions, the architecture will incorporate several strategies to maintain population diversity and encourage exploration.[19]

- **Maintaining Population Diversity:** The system will actively monitor the genetic diversity of the agent population. If diversity drops below a certain threshold, mechanisms such as introducing a batch of new, randomly generated "immigrant" agents can be triggered to inject fresh genetic material into the pool.[19]
- **Dynamic Mutation Rates:** The rate of mutation will not be static. If the average fitness of the population begins to plateau, indicating potential convergence on a local optimum, the mutation rate can be temporarily increased. This "triggered hypermutation" encourages more radical exploration, increasing the chances that an agent will make a leap across a "fitness valley" to a new, more promising region of the solution space.[19]
- **Non-Elitism:** While it is common to always preserve the best-performing individuals ("elites") from one generation to the next, this can accelerate convergence on local optima. ELSA will employ a mixed strategy that includes periodic non-elitism. In a non-elitist step, the best agents are not automatically carried over, giving novel offspring—even those with slightly lower initial fitness—a chance to survive and develop. This is critical for allowing the population to explore less-fit intermediate states that may be necessary stepping stones to reach a globally optimal solution.[24]

The use of a multi-objective fitness function will naturally drive the co-evolution of different agent archetypes within the swarm. The complex fitness landscape created by competing

objectives (e.g., performance vs. efficiency) will have multiple distinct peaks. An agent that evolves to maximize performance at any cost will occupy one peak, becoming a hyper-specialized "expert." Another agent that evolves a balanced profile, achieving good-enough results across all objectives, will occupy a different peak, becoming a "generalist." A properly managed selection process, using the diversity-preserving mechanisms described above, will not converge on a single winning strategy. Instead, it will maintain a population distributed across these various strategic niches. This creates a resilient internal ecosystem where the system can deploy specialists for critical, high-stakes tasks and generalists for broad, exploratory problems, making the entire swarm far more versatile and adaptive than a homogeneous population could ever be.

Part III: The Adaptive Mechanism - Goal-Directedness through Context-Aware Reinforcement Learning

The Genetic Algorithm described in Part II provides a powerful mechanism for "slow" evolution, optimizing the fundamental design of agents over many generations. To complement this, ELSA requires a "fast" adaptation layer that allows the system to modify its behavior in real-time, within a single task or conversation. This section details this adaptive mechanism, which uses Reinforcement Learning (RL) to instill goal-directedness and context-awareness in the agent swarm. This layer is what transforms ELSA from a system that merely self-improves over the long term into a truly self-adaptive engine that responds intelligently to its immediate environment.

3.1. Overcoming the Lack of Inherent Goal-Directedness in LLMs

A fundamental limitation of current LLMs is that they are not inherently goal-directed agents. They are, at their core, sophisticated sequence predictors that react to the immediate context of a prompt.[25] While they possess vast capabilities, they often fail to make full use of them to purposefully pursue a high-level goal over multiple steps or turns of interaction.[25] To build a truly autonomous system, an explicit mechanism for instilling and maintaining goal-directed behavior is required. Reinforcement Learning is the ideal framework for this purpose. It provides a mathematical foundation for teaching an agent a "policy"—a strategy for selecting actions in a given state—in order to maximize a cumulative reward signal over time, effectively teaching it to pursue a long-term objective.[27]

3.2. The RL Framework for Agent Adaptation

Each agent within the ELSA swarm will be equipped with an RL-based adaptive layer. This layer allows an agent to make subtle, on-the-fly adjustments to its own operational strategy based on the immediate context and feedback it receives.

- **State Space:** The "state" is the agent's perception of the environment at a given moment. For this mechanism to be effective, the state representation must be rich and highly contextual. It will incorporate multiple streams of information, leveraging principles from context-aware AI to ensure decisions are relevant and personalized.[29] The state will include:
  - The user's query, including the full history of the current interaction.
  - The current state of the shared stigmergic workspace, providing an objective view of the task's progress.
  - The operational status and recent outputs of other agents in the swarm.
  - Real-time performance metrics, such as response latency or resource consumption.
- **Action Space:** An agent's "actions" in this RL framework are not external manipulations but internal, behavioral modifications. This is a form of dynamic "meta-prompting" or self-rewriting, where the agent adjusts its own instructions for the current task.[27] The action space will consist of a set of discrete choices, such as:
  - **Constraint Modification:** Temporarily adding, removing, or rephrasing a specific constraint within its own prompt (e.g., "Be more concise," "Provide three distinct examples").
  - **Information Seeking:** Actively deciding to request clarification from the user or query another specialized agent for missing information.
  - **Parameter Adjustment:** Modifying its own LLM inference parameters for the next generation step (e.g., increasing temperature for more creative exploration, decreasing it for more precise execution).
- **Reward Model:** The reward model is the heart of the RL loop, providing the feedback signal that guides the learning process. A carefully designed reward model is essential to prevent "reward hacking," where an agent learns to exploit loopholes to get a high score for undesirable behavior.[9] The reward signal will be a composite function derived from multiple sources:
  - **Explicit Feedback:** Direct signals from the user, such as ratings or corrections.
  - **Implicit Feedback:** Inferred signals, such as the user accepting a suggestion or ending the conversation.
  - **Task-Based Metrics:** Objective measures of success, such as the F1 score for classification tasks or an exact match score for question-answering.[27]
  - **Internal Peer Review:** Feedback from Critic agents within the swarm, which can provide nuanced, real-time evaluation of an output's quality.

- **Safety Penalties:** Strong negative rewards for any action or output that is flagged as unsafe, biased, or misaligned by the Guardian agent swarm (Part V).

This two-tiered learning system, combining a slow, generational GA with a fast, real-time RL mechanism, is analogous to the interplay between genetics and learning in biological organisms. The Genetic Algorithm modifies the agent's "genotype"—its fundamental, inherited prompt-DNA that defines its potential. The Reinforcement Learning mechanism, in contrast, modifies the agent's "phenotype"—its expressed behavior in a specific context. An organism's genes define its innate capabilities, but its learned behaviors determine its success in a particular environment. This dual-layered approach provides ELSA with both long-term structural stability and optimization (from the GA) and short-term behavioral flexibility and responsiveness (from RL), creating a complete architecture that is both robust and adaptive.

3.3. Balancing Exploration and Exploitation

A central challenge in RL is the exploration-exploitation dilemma.[28] The agent must decide whether to "exploit" actions that have yielded high rewards in the past or to "explore" new, untried actions that might lead to even better outcomes. An agent that only exploits will become rigid and fail to discover better strategies, while an agent that only explores will perform erratically. To manage this trade-off, ELSA's agents will employ strategies such as an epsilon-greedy policy, where they choose the known-best action most of the time but select a random action with a small probability (epsilon), ensuring a constant level of exploration.

3.4. Adaptive Learning and Real-Time Processing

The RL mechanism enables true adaptive learning by allowing agents to update their policies based on the real-time flow of information.[29] If a particular self-modification (an "action") consistently leads to negative feedback from Critic agents or the user, the agent's policy will quickly learn to avoid that action in similar future states. Conversely, actions that lead to positive rewards will be reinforced. This continuous, low-latency feedback loop makes the entire system highly responsive to the nuances of a specific task, the evolving needs of the user, and the dynamic behavior of other agents in the swarm.

The reward model itself cannot be a static, objective oracle; it is an artifact that must co-evolve with the system's capabilities. A fixed reward model inevitably becomes a vulnerability. As LLMs are trained to maximize a score, they will eventually discover and exploit any flaws or loopholes in the reward model's logic, a phenomenon known as reward hacking.[9]

This represents a critical alignment failure. Therefore, the reward model cannot exist outside the evolutionary ecosystem. It must be subject to the same adversarial pressures as the rest of the system. The "safe-by-coevolution" paradigm posits that safety must evolve in lockstep with capability.[32] Applying this principle to the RL mechanism means the reward model—the very definition of "good" behavior—must be continuously red-teamed and refined. The Guardian agents (Part V) will be tasked with adversarially probing the reward model for exploits. The overall fitness of the ELSA system will then depend not just on its ability to achieve a high score from a given reward model, but on its ability to evolve a reward model that is itself robust and difficult to exploit. This transforms the problem of alignment from hitting a fixed target to engaging in a continuous, adversarial process of value definition.

Part IV: The Meta-Cognitive Core - A Self-Analysis and Semantic Fidelity Protocol

To achieve true autonomy and long-term stability, a self-evolving system must possess a degree of self-awareness. It needs the capacity to observe, record, and reason about its own developmental trajectory. This section details the most novel component of the ELSA architecture: its meta-cognitive core. This layer, managed by a dedicated class of Observer agents, provides the system with an "inner eye," enabling it to perform self-analysis, combat the subtle decay of meaning that plagues self-referential systems, and achieve an unprecedented level of interpretability.

4.1. The Observer Agent Swarm

A specialized subset of agents, the "Observers," form the meta-cognitive core of ELSA. To ensure their objectivity, these agents are functionally firewalled from the primary task-solving swarm. Their prompts, tools, and reward signals are distinct and focused solely on the task of monitoring and analysis. The primary function of the Observer swarm is to create and maintain a detailed, immutable "evolutionary ledger." This ledger serves as the system's long-term memory, meticulously logging every significant event in its lifecycle:

- **Genetic Modifications:** Every mutation and crossover event from the GA is recorded, detailing the specific change made to an agent's chromosome.
- **Fitness and Behavioral Impact:** The resulting change in the agent's multi-objective fitness score and its observable behavior is logged and linked to the genetic modification that caused it.

- **Adaptive Policy Shifts:** Significant changes in an agent's policy, as learned through the RL mechanism, are captured, providing a record of short-term adaptations.
- **Contextual Data:** The environmental context in which these changes occurred—including the type of task, user inputs, and the state of the swarm—is recorded to enable future causal analysis.

4.2. The Semantic Drift Protocol (SDP)

A critical and proactive function of the Observer swarm is the implementation of a Semantic Drift Protocol (SDP). This protocol is a direct countermeasure to the insidious problem of semantic drift, where the meaning and nuance of concepts erode over repeated generations of self-referential processing.[13] This phenomenon is a significant risk, as outputs can remain factually correct on the surface while their deeper purpose or intent becomes "hollowed out".[14]

The SDP is inspired by the "Telephone Game" evaluation protocol used in multi-modal research.[15] Periodically, an Observer agent will extract a core concept or foundational instruction from ELSA's system architecture (e.g., a key principle from a Critic agent's prompt). It will then subject this concept to a chain of transformations using the current generation of task-solving agents. A typical chain might be: original concept -> summarize -> rephrase for a technical audience -> explain the core trade-offs -> generate a logical formalism -> regenerate the original concept from the formalism.

The Observer agent then calculates a **Semantic Fidelity Score** by measuring the semantic distance between the initial concept and the final regenerated concept. This is accomplished using sophisticated embedding-based similarity metrics (such as those from CLIP or MPNet) that can capture nuances of meaning beyond simple keyword matching.[15] A high distance score indicates significant semantic drift. This score is then fed back into the system's learning mechanisms as a powerful corrective signal:

- In the GA, a high semantic drift score contributes a significant penalty to the swarm's overall fitness, creating direct evolutionary pressure against agents and configurations that degrade meaning.
- In the RL loop, actions that lead to outputs with low semantic fidelity receive a negative reward, teaching agents to favor behaviors that preserve conceptual integrity.

4.3. Causal Analysis and Evolutionary Rationale

The evolutionary ledger is more than a simple log; it is a rich dataset ripe for causal inference. The Observer swarm is periodically tasked with analyzing this data to generate hypotheses and narratives about the system's own evolution. Using prompts designed for meta-analysis, an Observer agent might produce an output like:

> "Analysis of Generations 150-200: The introduction of 'Gene Cluster 7B' (a prompt fragment emphasizing verifiable evidence) in Critic agents correlated with a 15% increase in the 'Robustness' fitness objective and a 5% decrease in the 'Efficiency' objective. This suggests a successful trade-off, where increased computational cost for fact-checking led to a more resilient system against adversarial inputs containing subtle misinformation. This evolutionary path was favored during a period where the Guardian swarm was deploying a high volume of factuality-based red-teaming attacks."

This self-generated analysis provides an unparalleled level of interpretability, allowing developers and the system itself to understand the *why* behind its evolution, not just the *what*.

The evolutionary ledger created by the Observer agents functions as more than a passive log for human analysis; it becomes an active component of the system's own cognitive architecture—a form of long-term, institutional memory. The importance of memory for enabling agents to learn from past interactions and adjust their behavior is well-established in MAS research [4], and context-aware AI systems rely heavily on historical data to inform their present actions.[30] By making the evolutionary ledger a queryable resource for the task-solving swarm, the system can learn from its entire history, not just from its immediate RL feedback loop. When faced with a novel or particularly challenging task, an agent can query the ledger to retrieve precedents, analyzing how similar problems were approached in its evolutionary past and which strategies proved most successful. This transforms the self-analysis feature from a passive reporting tool into an active driver of a powerful meta-learning loop, allowing the system to learn how to learn.

4.4. Preventing Self-Referential Degradation

The risk that LLMs learning primarily from their own AI-generated content will lead to a gradual degradation of quality, a narrowing of diversity, and the entrenchment of biases is a significant concern for the long-term viability of autonomous AI systems.[22] The ELSA architecture employs a two-pronged defense against this mode of collapse. The SDP is the primary, active defense, directly penalizing the loss of semantic integrity. As a secondary, passive defense, the Observer agents will constantly monitor the outputs of the task-solving

swarm for statistical indicators of degradation, such as a decrease in lexical diversity, a reduction in conceptual novelty, or an increase in the repetition of certain patterns. If these metrics fall below a predefined threshold, it can trigger an intervention, such as automatically increasing the GA's mutation rate to inject more novelty or flagging the need for an infusion of fresh, human-generated data into the system's learning environment.

The Semantic Drift Protocol provides a crucial function beyond simply preventing language from becoming "hollowed out." It serves as a stability anchor for the entire goal-directed system. The goals that drive the RL mechanism are defined by language within prompts and reward models. As semantic drift research demonstrates, the meaning of this language can subtly but significantly erode in iterative, self-referential systems.[13] This creates a catastrophic failure mode: the system could be perfectly optimizing for a goal whose meaning has drifted far from the user's original intent. Surface-level performance metrics might remain high, but the system would be fundamentally misaligned. Because LLMs are not inherently goal-directed [25], the stability of the linguistic constructs that define their goals is paramount. The SDP acts as a continuous check on this stability. By ensuring that the core concepts related to the system's goals and values remain semantically stable over time, the protocol provides a foundational guarantee for the long-term alignment and reliability of the entire adaptive architecture.

Part V: Advanced Dynamics - Niche Construction and Co-Evolutionary Safety

The architecture described thus far details a system that can adapt to a predefined environment. The final and most advanced layer of the ELSA framework elevates it from a reactive to a proactive system—one that not only adapts to its environment but actively shapes it. This section introduces the biological concept of niche construction and integrates a state-of-the-art, dynamic safety paradigm to ensure that this powerful, world-altering capability remains robustly aligned with human values.

5.1. Niche Construction Theory in ELSA

In evolutionary biology, niche construction is the process by which organisms actively modify their environment, thereby altering the selection pressures that they and other organisms are subject to.[36] An earthworm, for example, changes the chemical and physical properties of the

soil it inhabits, creating an environment that favors its own survival. This legacy of environmental modification, known as "ecological inheritance," is a powerful evolutionary force.[36]

Applied to ELSA, niche construction represents the system's ability to become a co-creator of its own evolutionary landscape. This capability will manifest in several key ways:

- **Active Data Curation:** The system will analyze its own performance and identify gaps or biases in its training data. It can then generate specific requests for new information from external, human-verified sources, effectively guiding the expansion of its own knowledge base.
- **Synthetic Curriculum Generation:** ELSA can generate novel, synthetic data and problem sets to practice specific skills or explore known edge cases where it is weak. This allows it to build its own targeted training curriculum, accelerating its development in areas of need.
- **User Interaction Shaping:** The system can move beyond passively responding to user queries. It can learn to proactively ask clarifying questions, suggest more effective ways to frame a problem, or guide the user towards more productive modes of interaction, thereby improving the quality of its own inputs and shaping the collaborative environment.

This capacity for niche construction transforms ELSA from a passive learner into a self-directed agent that actively drives its own evolution.

5.2. The Imperative for Dynamic Safety: Beyond Static Guardrails

A system with the emergent capabilities and environmental-shaping power of ELSA cannot be effectively constrained by a static set of safety rules or guardrails. Such pre-defined constraints are inherently brittle; they are based on known risks and cannot anticipate the novel behaviors and unforeseen failure modes that will inevitably emerge from a complex, evolving system.[32] Safety cannot be a fixed boundary applied post-hoc; it must be a dynamic, adaptive capability that evolves in tandem with the system's performance. This is the central tenet of the "safe-by-coevolution" paradigm, a new approach to AI safety that treats safety as a learned, adversarial, and ongoing process.[32]

5.3. The Guardian Agent Swarm: An Adaptive Immune System

To implement co-evolutionary safety, the ELSA architecture includes a second, distinct swarm

of agents: the "Guardians." The Guardian swarm co-evolves in parallel with the primary task-solving swarm. Its sole purpose is to ensure the safety, robustness, and alignment of the entire ELSA system. The Guardians use the same GA/RL evolutionary mechanisms as the main swarm, but they are optimized against a different fitness function—one that rewards the discovery and neutralization of risks.

The Guardian swarm is composed of specialized agent profiles analogous to the different components of a biological immune system:

- **Red-Teamers:** These agents evolve to become expert adversaries. Their goal is to craft prompts, inputs, and interaction sequences designed to break the main swarm's safety protocols, elicit harmful or biased outputs, or cause it to deviate from its intended goals.
- **Value Verifiers:** These agents check the outputs of the main swarm for alignment with a dynamic, multi-layered ethical framework. This framework can include core constitutional principles, societal norms, and user-defined values, allowing for a nuanced and context-aware form of value alignment.
- **Reward Model Auditors:** A highly specialized class of Red-Teamer that focuses exclusively on finding and exploiting loopholes in the RL reward model. Their success in "hacking" the reward model directly penalizes the system's fitness, forcing the reward model itself to evolve greater robustness.[9]
- **Anomaly Detectors:** These agents monitor the behavior of the entire system for statistical irregularities or out-of-distribution activity. They act as an early warning system, flagging novel or emergent behaviors that may not violate a specific, pre-defined rule but are nonetheless indicative of a potential new failure mode.[9]

5.4. The Co-Evolutionary Dynamic

The relationship between the task-solving swarm and the Guardian swarm is simultaneously adversarial and symbiotic. It creates a perpetual "arms race" that drives continuous improvement in both capability and safety. The dynamic works as follows:

1. The Guardian swarm's fitness is rewarded for successfully identifying a vulnerability (e.g., a Red-Teamer elicits a harmful response). This creates evolutionary pressure for the Guardians to become ever more sophisticated and creative in their attacks.
2. When a Guardian successfully finds a failure, the fitness of the main swarm is heavily penalized. This creates direct evolutionary pressure on the task-solving agents to develop robust defenses and become more intrinsically aligned.

This dynamic ensures that safety is not a one-time achievement but a constantly evolving property of the system. It is a direct implementation of the "AI-45° Law," which posits that for safe progress towards advanced AI, safety capabilities must advance along a 45-degree

trajectory with performance capabilities—neither can be allowed to lag behind the other.[32]

5.5. Co-Alignment: Bidirectional Human-AI Adaptation

The ultimate vision for this co-evolutionary framework extends beyond the AI system itself to include the human user. The goal is not simply AI alignment, where the system conforms to a fixed set of human preferences, but "Co-Alignment," a state of bidirectional human-AI cognitive adaptation.[43] In this paradigm, the human and AI mutually adapt to one another to achieve a state of symbiotic collaboration.[45] ELSA's niche construction capabilities are key to this process. By learning to guide users toward more effective prompting strategies, the AI can "teach" its human collaborators how to work with it more effectively. Simultaneously, the analyses produced by the Guardian and Observer swarms can highlight ambiguities or contradictions in human-stated values, prompting humans to reflect on and clarify their own intentions. This transforms the human-AI relationship from one of master and tool to one of co-evolving partners in a shared cognitive ecosystem.

Table V.1: Co-Evolutionary Safety Mechanisms

The following table maps the primary risks associated with a self-evolving system to the specific architectural mitigations and Guardian agent functions designed to counter them.

Identified Risk	Description	Architectural Mitigation	Guardian Agent Function	Supporting Research
Semantic Collapse / Model Collapse	The system learns from its own increasingly narrow outputs, leading to a loss of diversity,	Semantic Drift Protocol (SDP); Diversity Monitoring by Observer Agents.	**Semantic Fidelity Verifiers:** Continuously run the SDP and penalize the main swarm's fitness for	[14]

	nuance, and factual integrity over many generations.		drift. **Diversity Auditors:** Monitor output statistics and trigger interventions if diversity drops.	
Reward Hacking	The RL system finds and exploits a loophole in the reward model to achieve a high score for behavior that is misaligned with the intended goal.	Adversarial Co-evolution of the Reward Model.	**Reward Model Auditors:** Evolve to find and publicize exploits in the current reward model, forcing the model to be patched and become more robust.	9
Emergent Misalignment	Novel interactions between agents or unforeseen environmental factors lead to harmful collective behavior that was not explicitly programmed or forbidden.	Anomaly Detection; Continual Red-Teaming.	**Anomaly Detectors:** Monitor swarm behavior for statistical irregularities and flag novel patterns. **Red-Teamers:** Evolve novel multi-agent attack strategies to uncover emergent vulnerabilities.	9
Adversarial Vulnerability	The system is susceptible to	Distributed Defense via	**Red-Teamers:** Evolve to	9

	crafted inputs designed to deceive it, bypass safety filters, or elicit unintended and harmful actions.	MAS; Adversarial Co-evolution.	generate a continuous stream of novel adversarial attacks, forcing the main swarm to develop more generalizable and robust defenses.	
Alignment Collapse in Conversation	Over prolonged, unguided multi-agent interactions, the collective ethical alignment of the swarm degrades, leading to unsafe outputs.	Structured Debate Protocols; Value-based Criticism.	**Value Verifiers:** Act as participants or monitors in debates, checking outputs against a dynamic ethical framework and penalizing misaligned contributions.	10
Goal Drift	The system's internal representation of its core goals degrades semantically over time, causing its behavior to diverge from the user's original intent.	Semantic Drift Protocol (SDP) applied to core goal concepts.	**Goal Integrity Verifiers:** A specialized class of Semantic Fidelity Verifiers that focus exclusively on the stability of concepts within the fitness	14

			function and reward model.	

Part VI: Architectural Synthesis and Implementation Roadmap

This final section synthesizes the preceding components—the multi-agent foundation, the evolutionary and adaptive mechanisms, the meta-cognitive core, and the co-evolutionary safety framework—into a unified architectural vision. It provides a high-level conceptual diagram of the complete ELSA system and outlines a practical, phased roadmap for its research and development.

6.1. The Unified ELSA Architecture Diagram

The complete ELSA architecture can be visualized as a dynamic, two-tiered ecosystem.

- **The Core Interaction Hub:** At the center is the **Stigmergic Workspace**, a shared, structured environment that serves as the single source of truth for the current task state. This is the primary medium through which most agents interact.
- **The Task-Solving Swarm (Tier 1):** This swarm directly interacts with the workspace to solve user-defined tasks. It comprises the **Generator**, **Critic**, and **Refiner** agents, with the **Orchestrator** agent facilitating the flow of information between them and the workspace. This entire swarm is subject to two nested learning loops:
    - **The Inner RL Loop (Fast Adaptation):** Operates on a per-task, per-interaction basis. Agents receive reward signals based on performance and feedback, allowing them to make real-time adjustments to their behavior.
    - **The Outer GA Loop (Slow Evolution):** Operates across generations of the entire swarm. The fitness of each agent is evaluated, and the operators of selection, crossover, and mutation are applied to produce the next generation of agents with improved fundamental designs.
- **The Meta-Cognitive & Safety Swarm (Tier 2):** This tier operates in parallel, observing and interacting with the Tier 1 swarm.
    - The **Observer Agent Swarm** reads from the entire system—the GA, the RL loop, and the workspace—to populate the **Evolutionary Ledger**. It runs the Semantic Drift Protocol and provides feedback (semantic fidelity scores, diversity metrics) into the GA's fitness function and the RL reward model.

- The **Guardian Agent Swarm** acts as the system's adaptive immune system. It launches adversarial probes against the Task-Solving Swarm and the RL Reward Model. Its findings (successful attacks, identified vulnerabilities) also feed directly into the fitness and reward functions as strong negative penalties, driving the co-evolution of safety and capability.

This unified architecture creates a self-correcting, self-improving, and self-aware system where performance, safety, and interpretability are not separate features but emergent properties of the same dynamic, co-evolutionary process.

6.2. Phased Implementation Roadmap

The development of the full ELSA architecture is a significant, long-term research endeavor. A phased implementation approach is recommended to manage complexity and validate each component iteratively.

- **Phase 1: Foundational MAS.** The initial focus should be on implementing a static, non-evolving multi-agent system. This involves defining the initial profiles for Generator, Critic, and Orchestrator agents and establishing robust stigmergic and debate-based communication protocols. The primary goal of this phase is to demonstrate superior performance and robustness on complex tasks compared to a monolithic prompt architecture.
- **Phase 2: Introducing Real-time Adaptation.** Integrate the context-aware RL mechanism into the static MAS. This allows individual agents to perform intra-task self-modification. Development in this phase will center on creating a robust, multi-faceted reward model and demonstrating that agents can learn to adapt their behavior in response to real-time feedback.
- **Phase 3: Enabling Evolution.** Implement the GA engine and the Observer agent swarm. This phase involves defining the genetic representation of agents and the multi-objective fitness function. The system will begin to evolve agent prompts and configurations over generations. The initial version of the Semantic Drift Protocol should be integrated here to provide a crucial stability-oriented selection pressure from the outset.
- **Phase 4: Activating Co-Evolutionary Safety.** Deploy the Guardian agent swarm. This is a critical transition point where the system's safety mechanisms become dynamic. The co-evolutionary "arms race" between the Task-Solving and Guardian swarms begins, with the goal of demonstrating that safety capabilities can improve in lockstep with performance.
- **Phase 5: Unleashing Niche Construction.** In the final phase, the system is cautiously granted the ability to modify its own data environment and shape user interactions. This capability should only be enabled once the Guardian swarm has reached a high level of

maturity and has demonstrated its ability to effectively monitor and constrain the system's behavior in novel situations.

6.3. Open Challenges and Future Research Directions

The realization of the ELSA architecture presents several frontier research challenges that will require sustained investigation:

- **Computational Cost:** Simulating, evaluating, and evolving a population of thousands of LLM-based agents represents a formidable computational and engineering challenge. Research into more efficient simulation techniques, proxy models for fitness evaluation, and optimized MAS frameworks will be essential.
- **Reward Model Specification:** While the co-evolutionary approach makes the reward model more robust, the initial specification of the reward function remains a critical and difficult alignment problem. How do we encode complex human values into a reward signal without creating unforeseen loopholes?
- **Controlling Emergence:** The architecture is designed to produce beneficial emergent behavior. However, by its nature, emergence is difficult to predict. A key research area will be developing theoretical and empirical methods for understanding and bounding the emergent properties of complex AI ecosystems to ensure they remain beneficial.
- **Scalable Oversight:** As the system grows in autonomy and complexity, how can human oversight be maintained effectively? The goal is to create a system that is trustable without requiring constant, manual human intervention for every decision. Research into new models of human-AI collaboration, such as "constitutional AI" and scalable oversight techniques, will be paramount.

6.4. Concluding Vision: Towards Artificial Collective Intelligence

The ELSA architecture represents a paradigm shift in the design of intelligent systems. It moves away from the goal of creating a single, monolithic artificial "mind" and towards the cultivation of entire artificial "ecosystems." In this vision, intelligence is not a property of any single agent but a collective phenomenon that emerges from the interactions of a diverse, specialized, and co-evolving population. The ultimate objective is not merely a more capable or efficient AI model, but a new form of resilient, adaptive, and verifiably safe artificial collective intelligence that can enter into a state of co-alignment with humanity to solve our most complex challenges.

Works cited

1. Large Language Model Based Multi- agents: A Survey of Progress and Challenges - KAUST Repository, accessed October 11, 2025, https://repository.kaust.edu.sa/server/api/core/bitstreams/314fe81f-cfa1-4d93-8262-c1cde761c754/content
2. Large Language Model based Multi-Agents: A Survey of Progress and Challenges - arXiv, accessed October 11, 2025, https://arxiv.org/abs/2402.01680
3. Synergistic Simulations: Multi-Agent Problem Solving with Large Language Models - arXiv, accessed October 11, 2025, https://arxiv.org/pdf/2409.13753
4. Large Language Model based Multi-Agents: A Survey of Progress and Challenges - arXiv, accessed October 11, 2025, https://arxiv.org/html/2402.01680v2
5. Large Language Model Applied in Multi-agent SystemA Survey - ResearchGate, accessed October 11, 2025, https://www.researchgate.net/publication/386501622_Large_Language_Model_Applied_in_Multi-agent_SystemA_Survey
6. Definition and principles of swarm intelligence | Swarm ... - Fiveable, accessed October 11, 2025, https://fiveable.me/swarm-intelligence-and-robotics/unit-1/definition-principles-swarm-intelligence/study-guide/QmkQBeEQnvs1olWD
7. What are the key principles of swarm intelligence? - Milvus, accessed October 11, 2025, https://milvus.io/ai-quick-reference/what-are-the-key-principles-of-swarm-intelligence
8. AI Algorithms and Swarm Intelligence - Unaligned Newsletter, accessed October 11, 2025, https://www.unaligned.io/p/ai-algorithms-and-swarm-intelligence
9. AI safety - Wikipedia, accessed October 11, 2025, https://en.wikipedia.org/wiki/AI_safety
10. [Literature Review] Multi-Agent Large Language Models for Conversational Task-Solving, accessed October 11, 2025, https://www.themoonlight.io/en/review/multi-agent-large-language-models-for-conversational-task-solving
11. MALLM: Multi-Agent Large Language Models Framework - arXiv, accessed October 11, 2025, https://arxiv.org/html/2509.11656v2
12. Multi-Agent Large Language Models for Conversational Task-Solving - arXiv, accessed October 11, 2025, https://arxiv.org/html/2410.22932v2
13. Multi-Turn Semantic Drift - Arize AI, accessed October 11, 2025, https://arize.com/glossary/multi-turn-semantic-drift/
14. Semantic Drift: A Hidden Failure Mode in LLMs? : r/LLM - Reddit, accessed October 11, 2025, https://www.reddit.com/r/LLM/comments/1mxu4la/semantic_drift_a_hidden_failure_mode_in_llms/
15. The Telephone Game: Evaluating Semantic Drift in Unified Models - arXiv, accessed October 11, 2025, https://arxiv.org/html/2509.04438v2
16. Voting or Consensus? Decision-Making in Multi-Agent Debate - arXiv, accessed

October 11, 2025, https://arxiv.org/html/2502.19130v4
17. [2410.22932] Multi-Agent Large Language Models for Conversational Task-Solving - arXiv, accessed October 11, 2025, https://arxiv.org/abs/2410.22932
18. Multi-Agent Large Language Models for Conversational Task-Solving - arXiv, accessed October 11, 2025, https://arxiv.org/html/2410.22932v1
19. Genetic algorithm - Wikipedia, accessed October 11, 2025, https://en.wikipedia.org/wiki/Genetic_algorithm
20. [D] Why does the genetic algorithm tend to NOT produce garbage results? - Reddit, accessed October 11, 2025, https://www.reddit.com/r/MachineLearning/comments/onteod/d_why_does_the_genetic_algorithm_tend_to_not/
21. Genetic algorithms are not global optimization methods : r/MachineLearning - Reddit, accessed October 11, 2025, https://www.reddit.com/r/MachineLearning/comments/400kxn/genetic_algorithms_are_not_global_optimization/
22. Large Language Models and User Trust: Consequence of Self ..., accessed October 11, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11082730/
23. Large Language Models and User Trust: Consequence of Self-Referential Learning Loop and the Deskilling of Health Care Professionals - PubMed, accessed October 11, 2025, https://pubmed.ncbi.nlm.nih.gov/38662419/
24. Escaping Local Optima with Non-Elitist Evolutionary Algorithms - The Association for the Advancement of Artificial Intelligence, accessed October 11, 2025, https://cdn.aaai.org/ojs/17457/17457-13-20951-1-2-20210518.pdf
25. EVALUATING THE GOAL-DIRECTEDNESS OF ... - OpenReview, accessed October 11, 2025, https://openreview.net/pdf?id=BECkhjcofz
26. (PDF) Evaluating the Goal-Directedness of Large Language Models - ResearchGate, accessed October 11, 2025, https://www.researchgate.net/publication/390846019_Evaluating_the_Goal-Directedness_of_Large_Language_Models
27. Using Reinforcement Learning and LLMs to Optimize Prompts, accessed October 11, 2025, https://www.prompthub.us/blog/using-reinforcement-learning-and-llms-to-optimize-prompts
28. Reinforcement Learning Explained : r/PromptEngineering - Reddit, accessed October 11, 2025, https://www.reddit.com/r/PromptEngineering/comments/1igpml9/reinforcement_learning_explained/
29. What is contextual AI? - WalkMe - Digital Adoption Platform, accessed October 11, 2025, https://www.walkme.com/glossary/contextual-ai/
30. What is Context Awareness in AI? - Experience Dynamics, accessed October 11, 2025, https://frankspillers.com/what-is-context-awareness-in-ai/
31. How Context-Aware AI Enhances Customer Calls - Dialzara, accessed October 11, 2025, https://dialzara.com/blog/how-context-aware-ai-enhances-customer-calls
32. arxiv.org, accessed October 11, 2025, https://arxiv.org/html/2509.06786v1

33. What is context awareness in AI? - Graphite, accessed October 11, 2025, https://graphite.dev/guides/context-awareness-in-ai
34. (PDF) Large Language Models and User Trust: Consequence of Self-Referential Learning Loop and the Deskilling of Health Care Professionals - ResearchGate, accessed October 11, 2025, https://www.researchgate.net/publication/379123584_Large_Language_Models_and_User_Trust_Consequence_of_Self-Referential_Learning_Loop_and_the_Deskilling_of_Healthcare_Professionals
35. [2403.14691] Large Language Models and User Trust: Consequence of Self-Referential Learning Loop and the Deskilling of Healthcare Professionals - arXiv, accessed October 11, 2025, https://arxiv.org/abs/2403.14691
36. Evolution of Complex Niche-Constructing Behaviors and ... - Frontiers, accessed October 11, 2025, https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2020.00045/full
37. Niche Construction Theory: A Practical Guide for Ecologists | The Quarterly Review of Biology: Vol 88, No 1, accessed October 11, 2025, https://www.journals.uchicago.edu/doi/abs/10.1086/669266?mobileUi=0&journalCode=qrb
38. Evolutionary consequences of niche construction and their implications for ecology - PNAS, accessed October 11, 2025, https://www.pnas.org/doi/10.1073/pnas.96.18.10242
39. R^textbf{2AI}: Towards Resistant and Resilient AI in an Evolving World, accessed October 11, 2025, https://goatstack.ai/articles/2509.06786
40. R^textbf{2AI}: Towards Resistant and Resilient AI in an Evolving World - Hugging Face, accessed October 11, 2025, https://huggingface.co/papers/2509.06786
41. SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law | AI45 Research, accessed October 11, 2025, https://ai45.shlab.org.cn/research/posts/safework-r1/
42. SafeWork-R1: Coevolving Safety and Intelligence under the AI-45◦ Law - arXiv, accessed October 11, 2025, https://arxiv.org/html/2507.18576v1
43. Co-Alignment: Rethinking Alignment as Bidirectional Human ... - arXiv, accessed October 11, 2025, https://arxiv.org/abs/2509.12179
44. From AI Alignment to Recursive Alignment: Three Stages of Civilization-Safe Design | by nettalk83 | Aug, 2025 | Medium, accessed October 11, 2025, https://medium.com/@nettalk83/from-ai-alignment-to-recursive-alignment-three-stages-of-civilization-safe-design-adc920f9c9ec
45. Redefining Superalignment: From Weak-to-Strong Alignment to Human-AI Co-Alignment to Sustainable Symbiotic Society - arXiv, accessed October 11, 2025, https://arxiv.org/html/2504.17404v1
46. Survey of Multi-agent LLM Evaluations - LessWrong, accessed October 11, 2025, https://www.lesswrong.com/posts/tGcLA596E8g3KnphE/survey-of-multi-agent-llm-evaluations